

DriftLoc — Failure Analysis

Phase 2 navigation-error recovery (pose-variant localization + rejection)

1. What the system does

We take a high-mag DRAM reference crop and look for the same site in a 1000×1000 search image. Scale and rotation are unknown. Some pairs have no true match at all.

The pipeline is: multi-scale / multi-angle ZNCC proposals → small CNN verifier (hard negatives from other ZNCC peaks) → sub-pixel refine → local pose refine → confidence gate. If confidence is below 0.38 we output `found=0` and zero the pose columns.

Coarse-to-fine proposals keep median runtime around **0.58 s/pair**, well under the 5 s budget.

2. What we measured

Held-out organizer-style `phase2_test_set` (450 pairs: A 180 / B 180 / C 90), same path as `register.py`:

Check	Result
Rejection F1	0.965
Absent correct reject	77/90 (85.6%)
Present false reject	12/360 (3.3%)
Accepted A / B p@2	1.00 / 0.98
Median error (accepted)	0.3 px
Median $ \Delta\theta $ / $ \Delta\text{scale} $	0.09° / 0.50%

A noise-severity split on the same set is monotone: cleanest quartile p@1 = 0.98, harshest quartile p@1 = 0.72, but p@2 stays ≥ 0.98 . The one large miss (712 px) is in mid severity, not the noisiest bin — so it is a lattice error, not a noise error.

These numbers are on synthetic / organizer-style imagery, not production tool hardware.

3. Failure modes

Wrong DRAM cell (periodic structure)

Several neighbouring cells look almost the same, so ZNCC can peak on the wrong one. The verifier is trained on those look-alikes; near-tied scores go to the candidate closest to the image centre. This is rare on the test set (accepted worst case is usually a few pixels; one Set B pair jumped to hundreds of pixels). When it happens, localization and pose credit for that pair go to zero.

Noisy or sheared Set B images

Raster drift, charging, and dose noise weaken the match. We still find the right cell most of the time; the typical cost is sub-pixel → ~1-2 px, not a wrong site. Severity climbs smoothly. A few present pairs also drop below the reject gate.

Scale / rotation outside the search grid

We only search scale 8-12 and $\pm 5^\circ$. Local refine only polishes around the winner. If a pair sits outside those bounds, we never evaluate the true pose. That is an expected limit of the Phase 2 grid, not something we measured on this test set.

Reference actually absent (Set C)

The decoy is still real DRAM, so correlation can look decent. The gate uses verifier confidence, not raw ZNCC (ZNCC barely separates present from absent). We reject 85.6% of absent pairs. The remaining 13 false accepts are the precision side of the F1 trade-off. A few of those decoys have high confidence (one at 0.79); the verifier is sure, and wrong.

Many candidates with similar CNN scores

Repeated cells split probability, so top-1 mass can be low even when the crop is right. We only break ties within 0.05 of the best score, toward the image centre. This is why some present pairs look well-matched on ZNCC but still get `found=0`.

Sub-pixel / pose leftover

The grid steps are 0.5 in scale and 1°. We refine locally after that. Residual error is small when the cell is correct. We do not model full shear or barrel warp; that would be the next place pose would slip on a harder set.

Very weak / low-contrast reference

A flat ZNCC map gives few or no peaks. We then write `found=0`. We did not count these as a separate bucket on the test set.

4. What actually costs score

When we accept a pair, localization is already strong. The leftover risk is:

1. **False reject** — 12 present pairs thrown away (3.3%). Each loses localization and pose credit. Several of these still had high ZNCC; the softmax is a ranking score, not a true “is anything here?” detector.
2. **False accept** — 13 absent pairs (14.4%). They hurt rejection F1, which is already 0.965.
3. **Rare wrong-cell hits** — one large-error outlier, independent of noise level.

Raising or lowering the 0.38 gate only moves F1 around 0.960–0.968 on this set. We left it as trained. A dedicated present/absent head would be the real fix; we did not retrain before submit.

5. What we are not claiming

We are not claiming bit-identical behaviour on a noise model that is very different from training (we saw that on one extra generator). We are not claiming deformable registration. We are not claiming this was run on a real inspection tool.

6. Takeaway

The accept path is accurate across the severity range we could measure. The remaining mistakes are lattice ambiguity and using ranking confidence as a reject score. For this Phase 2 contract that trade-off is stable enough to ship.

Optional page-2 captions: green = GT box, red = prediction; one ≤ 2 px Set B residual; one false-reject where the peak looks right but `conf < 0.38`.